

Summary

- Develop machine learning methods for scientific discovery across geometric learning, molecular representation learning, and model-guided materials design, with publications at **ICML 2026** and **ICLR 2024**.
- Translate AI methods into industrial discovery workflows through **EXAONE Discovery** and programs that connect molecular screening with synthesis and experimental validation.
- Build optimization and scheduling systems for autonomous laboratories, combining constraint programming, online replanning, and robotic experiment execution.

Career Highlights

Scientific ML Research

Geometric transfer learning
Molecular representation learning
ICML 2026 / ICLR 2024

Industrial Deployment

EXAONE Discovery
Cosmetics / Rhamsydil
Battery electrolytes

Autonomous Systems

Robotic laboratory scheduling
Online replanning
Constraint optimization

Research Experience

LG AI Research

Research Scientist, Materials Intelligence Lab

Sep. 2022 – Present
Seoul, South Korea

- Developed **scientific ML methods** for molecular property prediction and transfer across tasks, including geometry-based models of latent manifolds and curvature-guided alignment.
- Built model-guided discovery pipelines across cosmetic ingredients, flame-retardant polymers, battery electrolyte solvents, and **EXAONE Discovery**.
- Connected AI model development with candidate selection, synthesis, and experimental validation, enabling industrial materials discovery programs with LG H&H, LG Chem, and LG Energy Solution.
- Developing **autonomous laboratory scheduling** and online replanning algorithms for *AI Autonomous Laboratory for Chemical Process and Materials Synthesis Optimization*.

LG AI Research

Research Intern, Materials Informatics Lab

Oct. 2021 – Feb. 2022
Seoul, South Korea

- Developed RL-based molecular optimization models for TADF materials in OLED discovery.

Selected Research & Industrial Impact

Geometric Representation Learning for Molecular Transfer Learning

Co-authored **GATE** and **GEAR**, extending transfer learning for molecular regression through differential geometry, global manifold alignment, and Ricci curvature matching. The resulting technology is used as one of the core technologies in **EXAONE Discovery**.

ICLR 2024 – ICML 2026

AI-Guided Materials Discovery Programs

Contributed to AI-driven discovery programs with LG H&H, LG Chem, and LG Energy Solution, spanning Rhamsydil, non-halogen flame retardants, and high-ionic-conductivity, low-gas electrolyte solvents.

2022 – Present

Autonomous Laboratory Scheduling & Optimization

Developing the scheduling engine for a fully closed-loop autonomous materials discovery system, focusing on constraint optimization, online replanning, resource allocation, and robotic experiment execution.

2025 – Present

Selected Publications

Geometric Embedding Alignment via Curvature Matching in Transfer Learning

2026 ICML

• Sung Moon Ko^{*}, Jaewan Lee^{*}, **Sumin Lee^{*}**, Soorin Yim, Sehui Han

From Ricci Curvature to Metric Matching: A Simplified Approach to Geometric Transfer Learning

2026 ICML AI4Science / AI4Math

• Sung Moon Ko^{*}, Jaewan Lee^{*}, **Sumin Lee^{*}**, Soorin Yim^{*}, Sehui Han^{*}

Geometrically Aligned Transfer Encoder for Inductive Transfer in Regression Tasks

2024 ICLR

• Sung Moon Ko^{*}, **Sumin Lee^{*}**, Dae-Woong Jeong^{*}, Woohyung Lim, Sehui Han

ArkDTA: Attention Regularization Guided by Non-Covalent Interactions for Explainable Drug-Target Binding Affinity Prediction

2023 Bioinformatics

• Mogan Gim, Junseok Choe, Seunghyun Baek, Jueon Park, Chaeun Lee, Minjae Ju, **Sumin Lee**, Jaewoo Kang

Improved Binding Affinity Prediction Using Non-Covalent Interactions and Graph Integration

2022 IEEE BigComp

• Junseok Choe, Keonwoo Kim, Minjae Ju, **Sumin Lee**, Jaewoo Kang

^{*} Equal contribution.

Education

Korea University	Sep. 2020 – Oct. 2022 Seoul, South Korea
M.S. in Computer Science and Engineering, Data Mining & Information Systems Lab Advisor: Prof. Jaewoo Kang	
Korea University	Mar. 2015 – Sep. 2020 Seoul, South Korea
B.S. in Environmental Science and Ecological Engineering; B.E. in Computer Science and Engineering	

Technical Expertise

Research Areas	Scientific machine learning, representation learning, geometric deep learning, transfer learning, molecular property prediction, scientific foundation models, autonomous scientific discovery
Optimization & Systems	Scheduling, constraint optimization, CP-SAT, online replanning, resource allocation, robotic laboratory workflows
Domains	Materials intelligence, polymers, electrolytes, cosmetics, molecular discovery, drug-target interaction modeling
Tools	Python, PyTorch, graph neural networks, RDKit, OR-Tools/CP-SAT, SQL, Linux
Languages	Korean (native), English (fluent professional proficiency; TOEIC 980)